

Permissively Licensed Rooftop Array Detection at 21 cm, with an Executable Gate

and Per-Array Roof Geometry from Public Lidar

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Abstract

Any analysis of rooftop solar that starts from overhead imagery inherits the error of its detector, and most of what that error costs is invisible in the headline metric. We report a rooftop array detector and a per-array roof-geometry method built entirely on public data and permissively licensed models, and evaluate each against evidence the model never saw. On 2025 Santa Cruz County orthoimagery at 0.204 m ground sample distance, RF-DETR reaches tile-level box- F_1 **0.826** (95% CI [0.798, 0.853], recall 0.803) under a gate stated as an executable script, with the confidence threshold frozen on validation and intervals bootstrapped over tiles rather than objects. Against county building permits the detector never trained on, parcel-matched recall is **73.6%** for installs the imagery could contain. A SAM2 mask stage makes area usable, and its one dangerous failure, segmenting the whole roof instead of the array, turns out to be a prompt-box problem: shrinking the prompt 15% moves median mask IoU from 0.509 to **0.844** and whole-roof grabs from 43.3% to **0%**. Plane fits to public 3DEP lidar recover tilt for 91.3% of arrays, with a median that matches 8,573 installer filings to within 0.03°, provided one projection trap is avoided: fitting on web-Mercator coordinates reports every tilt **18.9%** too shallow while producing a plausible distribution. We report each result with the evidence that bounds it.

1 Introduction

Programmes that target rooftop solar, whether for cleaning, maintenance, incentives or grid planning, need to know where the arrays are, how large they are and which way they face. Installers and utilities hold that information for their own customers; nobody holds it for a whole county in a form a municipality or a small organisation can use. Public aerial imagery and public lidar together could supply it, and learned detectors are good enough that locating arrays is no longer the hard part.

What remains hard is knowing how far to trust the result. Three gaps recur. First, a detector’s accuracy is usually reported as a number from a held-out split, scored in a way that is easy to make optimistic without noticing: thresholds

tuned on the test split, intervals computed over correlated objects, crops scored instead of the tiles the system actually processes. Second, downstream uses consume *area* and *orientation*, not detections, and the errors that matter for area, such as a mask spilling across a roof, are not visible in a box metric. Third, the widely used detection implementations are AGPL-3.0, which obstructs reuse by exactly the public bodies a public-data pipeline is for.

This paper reports a detection and geometry stack built to close those gaps, and it is the input stage of a companion analysis of rooftop soiling and cleaning economics.¹ Its contributions are:

1. **A detection gate stated as an executable definition** rather than a number, scored at the tile level through the production inference path, with the confidence threshold frozen on validation and a bootstrap over tiles (§3.1), and a paired comparison that detects a real improvement the marginal intervals hide (§4.1).
2. **An independent recall probe** against county building permits, matched by parcel rather than geocoded point, with the reasons the point method fails and the jurisdictional trap that can make permit coverage look like model error (§3.2).
3. **A mask stage whose failure mode is located in the prompt**, with two principled alternatives measured and rejected (§4.2).
4. **Per-array tilt and azimuth from public lidar**, a projection trap that biases tilt by 18.9% on web-Mercator point clouds, and a validation against installer filings that states what a distributional comparison can and cannot establish (§3.3, §4.3).

2 Related work

Rooftop PV detection.

Public rooftop-PV detection has an established lineage. The Duke dataset [2] released 0.3 m imagery with polygon

¹Both drafts are in the code repository: <https://github.com/CameronGordon/solar-soiling-detection>.

annotations for thousands of arrays, and DeepSolar [11] built a national deployment database by image classification with weak localisation. Both answer *whether* and *where*; neither was built to deliver per-array area for a downstream calculation. We use RF-DETR [9], a real-time variant of the detection transformer [4], with a SAM2 [5, 8] mask stage. All three are Apache-2.0. The YOLO implementations common in this literature are AGPL-3.0, and an earlier version of this pipeline used one, with slicing-aided inference [1] for small arrays; it is retained only as an evaluation baseline.

Roof geometry.

3D-PV-Locator [6] recovers tilt and azimuth per system from 3D building data and validates against a per-system registry, which is the stronger design where such a registry exists. We work from USGS 3DEP [10] point clouds, which cover much of the United States at no cost, and validate against installer-reported tilt from California interconnection filings [3]. Those filings carry no location finer than a service city, so our validation is distributional, and we say so where it matters (§4.3).

Evaluation practice.

Two protocol points in this paper are standard in principle and often skipped in practice: tuning an operating threshold on a split other than the one reported, and resampling the independent unit (here the tile) rather than correlated objects when computing intervals. We state both as parts of the gate rather than as analysis choices, so that they cannot be relaxed silently between versions.

3 Methods

3.1 Array detection

Imagery.

Santa Cruz County publishes 2025 aerial orthoimagery as a free map service. We tile a pilot area of interest over the Live Oak and Pleasure Point side of the city into 249 tiles covering 14.93 km² of *actually imaged* ground; the enclosing bounding box is 23.71 km², and the distinction matters for any recall denominator. True ground GSD is 0.204 m. Tiles are stored in EPSG:3857, where the nominal pixel size reads 0.256 m, inflated by $1/\cos(37^\circ)$. Each tile records its CRS and affine transform, so any pixel-space detection projects back to geographic coordinates unambiguously. The county service natively serves 6.3 cm and we under-requested by $3.3\times$ linear, so the ceiling on this method sits above what we report.

Labels and splits.

Arrays are annotated as polygons, because array extent feeds the area-to-dollar chain and a box overstates it on non-rectangular roofs. The dataset comprises 976 chips (680

train / 100 validation / 196 test) and 3,894 chip-level polygons, rebuilt from 244 of the 249 tiles. **Two ground-truth counts exist and are not interchangeable:** chip-level (validation 385, test 636) double-counts arrays that the ~ 80 -pixel chip overlap repeats and splits those the seams divide, whereas tile-level (validation 340, test 585) does not. The gate scores tile-level; a tile-level F_1 quoted against 636 is wrong by construction.

Model and inference path.

RF-DETR at 728-pixel input, single class. Production inference decomposes each full tile into a 2×2 grid of 640-pixel chips, runs one forward pass per chip, and merges across seams by non-maximum suppression at IoU 0.55.

The gate, stated as an executable definition.

Our shipping criterion is *whatever scripts/detect/eval_tile_f1.py prints*: tile-level detection F_1 at box-IoU ≥ 0.50 , micro-averaged over the split, measured through the production path above. Three protocol choices define it.

1. **Confidence is tuned on validation and frozen; test is the reported number.** Tuning confidence on test spends the held-out split. The cost of not doing so is measured rather than assumed: test’s own best F_1 is 0.8431 against 0.8260 at the validation-frozen threshold (Figure 1).
2. **Matching is on boxes, not masks**, so the gate is invariant to the presence of the mask stage. SAM2 changes area accuracy, not what was detected; conflate the two and a regression cannot be traced to its stage.
3. **The 95% interval bootstraps over tiles, not objects.** Arrays within a tile are correlated, and resampling objects yields an interval that is too narrow.

Passing requires all three of: $F_1 \geq 0.75$; F_1 95% CI lower bound ≥ 0.70 ; and recall ≥ 0.70 . Recall carries its own condition because a missed array is an unrecoverable missed lead, while the permit join filters false positives downstream.

Mask stage and prompt-box sensitivity.

SAM2 is in the product path not because the detector needs it, since the gate is box-based, but because *area is the product*. Prompted with the unmodified detector box, SAM2 returns whole-roof masks at a high rate. We treat the prompt box as the design variable and sweep it (§4.2).

3.2 Independent recall probe

Recall measured against held-out labels overstates recall against the real install base, because labels and model see the same imagery. We therefore cross-check detections against Santa Cruz County building-permit records, which the detector never trained on. Matching is by parcel APN rather

Table 1: Tile-level detection gate, $\text{box-IoU} \geq 0.50$, micro-averaged over the test split through the production chip-grid path. Confidence c^* tuned on validation and frozen. Intervals bootstrap over tiles ($n_{\text{tiles}} = 49$, $n_{\text{gt}} = 585$).

	W2 (shipped)	W1 (anchor)
c^* (val-frozen)	0.50	0.40
Precision	0.8499	0.8533
Recall	0.8034	0.7556
F_1	0.8260	0.8015
95% CI	[0.798, 0.853]	[0.773, 0.830]
Test-tuned best F_1	0.8431	0.8044
Gate verdict	PASS	PASS

than geocoded point. The Census batch geocoder interpolates each address onto the street segment, so points land a median 22 m from the nearest detected array and zero of 126 point matches fall inside a polygon. The point method measures geocode quality, not detection.

3.3 Roof geometry from public lidar

We fit a plane to USGS 3DEP [10] lidar returns inside each detected polygon, recovering per-array tilt and downslope azimuth over plain HTTP against the Entwine octree. The headline 3DEP product is a bare-earth DEM and is useless here by construction, since roofs are removed from it; the point cloud is not.

A projection trap.

Entwine X/Y are EPSG:3857 metres while Z is true metres. At this latitude the Mercator scale factor is $1/\cos(36.97^\circ) = 1.2515$, so a plane fitted on raw coordinates reports tilt **18.9% too shallow, silently, on every roof, producing an entirely plausible distribution**. We deflate X/Y by $\cos(\text{lat})$ and pin the magnitude in a unit test, so the correction cannot be removed unnoticed. Anyone reproducing this on web-Mercator lidar will hit it.

Failed fits are reported as null.

Arrays without a usable fit keep a flagged flat default and are never median-filled. Filling them would hand every unfitted roof the typical angle and quietly remove the variation a downstream analysis would test for.

4 Results

4.1 Detection

Table 1 reports the gate. The shipping checkpoint passes all three conditions. Figure 1 shows the full operating curve on both splits and the size of the tuning penalty.

Paired comparison, not overlapping intervals.

Doubling the training set moved F_1 from 0.8015 to 0.8260. The two marginal intervals overlap heavily and each point estimate lies inside the other’s interval, so the naive comparison reads “within noise.” That is the wrong test: both models were scored on the same 49 tiles, so the comparison is paired. Bootstrapping the per-tile difference gives $\Delta F_1 = +0.0246$, 95% CI $[+0.0051, +0.0441]$, $P(W2 > W1) = 0.993$, concentrated in recall $(+0.0479, \text{CI } [+0.0202, +0.0753])$, $P = 0.999$. The marginal comparison would have discarded a real gain.

Two caveats on the shipped checkpoint.

The tuning penalty grew relative to the anchor (0.017 against 0.003), so validation and test disagree more about where the operating point sits; and the chip-to-tile gap widened (0.029 against 0.008). Both are tracked and neither is yet explained.

Independent recall.

Against 434 geocoded permit points inside the imaged footprint, APN-matched recall is **55.3%** (240/434) across all years and **73.6%** (156/212) restricted to pre-cutoff installs that the imagery could contain. The point method reads 29.0% (Wilson CI 25.0–33.5%) for the geocoding reason given in §3.2. Three caveats travel with the number. Part of the gain could in principle come from spurious detections, since APN recall counts a parcel as found if any detected polygon’s interior point falls inside it; gate precision of 0.850 bounds but does not eliminate this. Permits are a valid recall probe and an *invalid* precision denominator: 85% of hand-labelled arrays sit on parcels with no permit on file, because the registry begins in 2016 and the residential boom preceded it. And 1,310 of 1,865 sites fall in city APN books with 0.0% county-permit coverage, because the City of Santa Cruz is a separate permitting authority. Any study joining detections to permits must check jurisdiction before reporting coverage.

4.2 Mask quality is a prompt-box problem

Figure 2 sweeps the prompt box over 60 validation arrays. Prompting with the raw detector box yields median IoU 0.509, median mask area $1.97\times$ ground truth, and a 43.3% rate of whole-roof grabs. Shrinking the box 15% yields median IoU **0.844**, median area ratio **1.01**, and **0.0%** roof-grab. Area is therefore essentially unbiased, which is exactly what the area-to-kW-to-dollar chain needs.

Two principled alternatives, both measured, both rejected.

Background negative-point prompts behave exactly as theory predicts: they help when the box is bad (roof-grab 38.3%→26.7% at dilate-50%) and hurt when it is good (0.844→0.820, introducing 1.7% roof-grab where there was none), because with an already-shrunk box the ring sits near

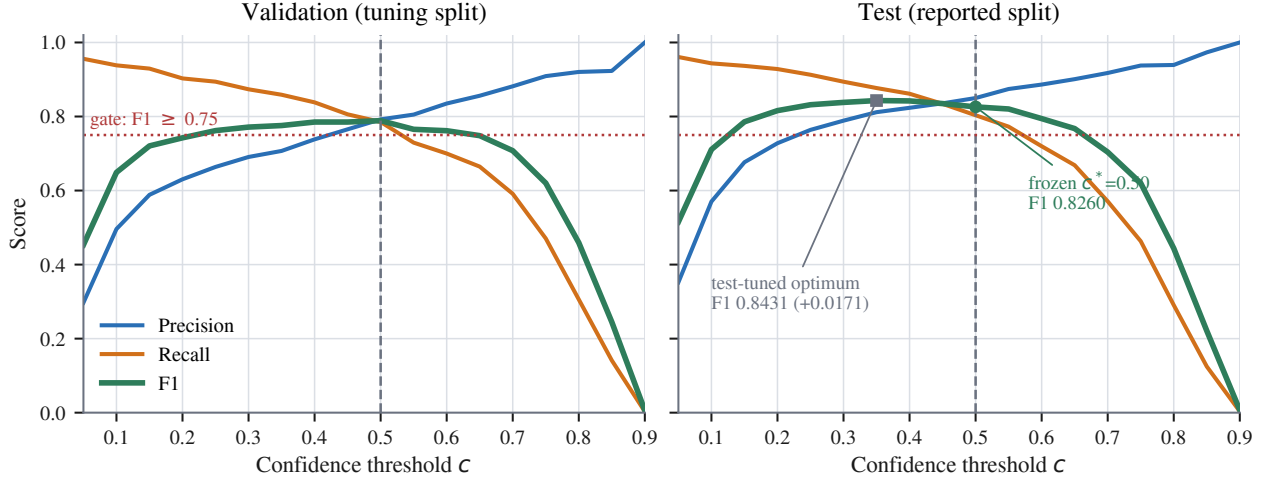


Figure 1: Operating curves on both splits. The frozen threshold $c^* = 0.50$ was chosen on validation alone; the test-tuned optimum sits 0.017 F_1 higher, which is the measured price of not spending the held-out split.

the true panel edge and clips real pixels. Multimask candidates clipped to the detector box with area-fit selection were worse everywhere (0.824 $\rightarrow$ 0.673 on exact boxes; roof-grab 1.7% $\rightarrow$ 71.7% at dilate-25%), because picking the candidate whose area matches the detector box amplifies box error in exactly the regime the change targeted.

A ceiling we cannot locate.

We never measured inter-annotator agreement, because our annotation platform deduplicates and one tile cannot go to two labellers. If two humans agree to only ~ 0.85 IoU on a 26-pixel array, then 0.844 is the label-noise floor and further mask work is unmeasurable. We do not know which side of that line we are on.

4.3 Roof geometry

Plane fits succeed for 3,068 of 3,362 arrays (91.3%), with median RMS residual 0.040 m over ~ 270 points per array; synthetic planes are recovered to $< 0.2^\circ$ tilt and $< 1^\circ$ azimuth. Figure 3 compares the recovered distribution against installer-reported tilt from 8,573 utility interconnection filings for the same county—a different instrument, a different decade, and no shared failure mode. **Both medians are 19.0° .** The low tails differ for a documented reason: the filings encode “not reported” as 0° and we drop those rows, so the filings understate the shallow tail (0.78% below 5°) while the lidar tail stays unbiased (8.38%). That 8.38% is directly relevant to Mejía and Kleissl [7]’s finding on sub-five-degree arrays.

What this comparison can and cannot establish.

It is distributional, not paired, and that is a property of the source rather than a choice: the DGStats interconnection extract carries a service city and nothing finer—no parcel num-

ber, address or coordinate—so no lidar array can be matched to the filing that describes it. 3D-PV-Locator [6] validates per system and is the stronger design where such a key exists.

We also decline to read agreement from the medians alone. A two-sample Kolmogorov–Smirnov test rejects decisively ($D = 0.214$, $p < 10^{-90}$, $n = 3,068$ against 8,573): the lidar distribution has much heavier tails, with a 10th percentile of 6.4° against 14° reported and a 90th of 30.8° against 27° . The mechanism is visible in the reported values. Every one is an integer, five values account for 57.6% of filings, and **27% are exactly 18°** —the 4:12 roof pitch. Installers file a nominal pitch from a small menu; the lidar fit measures a continuous angle on whatever roof is actually there. That the two medians land 0.03° apart across such different generating processes is the real result, and it is a statement about central tendency only. Per-array evidence, where we have it, comes instead from the standoff natural experiment below.

A natural experiment supports the interpretation: the lidar flight predates some installations and postdates others, and permit dates split the population on a variable unrelated to lidar. The panel standoff step appears exactly where physics says it should: pre-flight median $+0.145$ m against post-flight $+0.013$ m, difference $+0.132$ m, bootstrap 95% CI $[+0.084, +0.208]$, $p < 0.0001$. The fit therefore reports true racking angle where the array predates the flight and roof pitch where it does not. The groups are small ($n = 37$ and 54) and only 15.9% of arrays yield a usable standoff.

5 Discussion

What carries over to other detection work.

Most of what this paper contributes is protocol rather than architecture, and protocol is what transfers. Stating the

SAM2 prompt-box sensitivity (n=60 validation arrays, orange = production default)

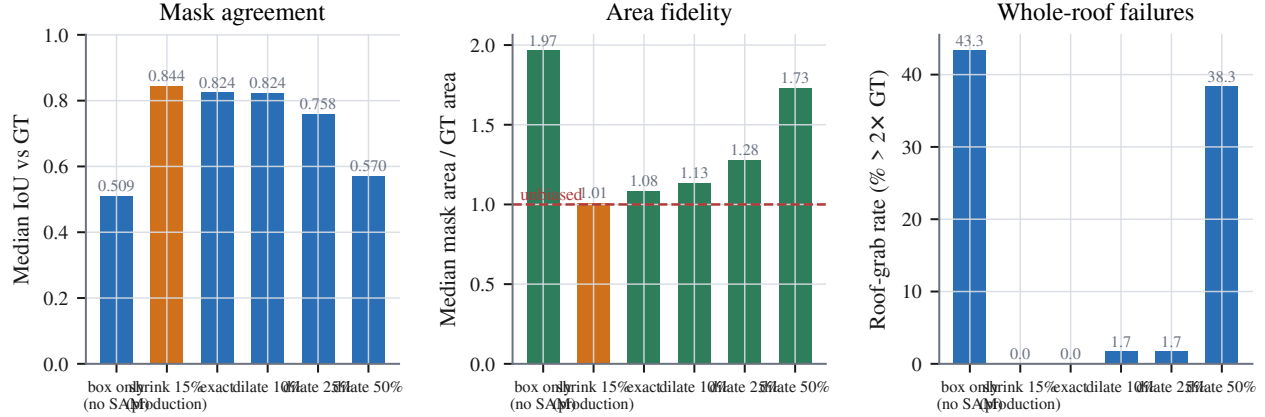


Figure 2: SAM2 prompt-box sensitivity. The whole-roof failure mode is created by an oversized prompt, not by the mask model, and disappears at the production default.

gate as a script means a later model is scored by the same code, through the same inference path, with the same frozen-threshold rule; a gate stated in prose drifts every time someone re-implements it. Scoring whole tiles rather than crops keeps the number tied to what the system actually does: the shipped model scores 0.029 F_1 lower at the tile level than on chips, a gap a crop-level number would have hidden. And the paired bootstrap turned an apparent tie into a measured gain. Reading overlapping marginal intervals as “no difference” would have discarded the effect of doubling the training set.

Treat the prompt as a design variable.

The mask result generalises beyond solar. When a promptable segmenter is fed boxes from an upstream detector, the box is an input with its own error distribution, and the cheapest fix for a segmentation failure may be to change the prompt rather than the mask model or its post-processing. Both of the more principled fixes we tried made results worse in the regime they targeted, for reasons that follow from how the prompt interacts with box error.

Validation that does not share a failure mode.

The two independent checks, permits for recall and installer filings for tilt, are useful precisely because they are generated by processes unrelated to the imagery. Each also came with a trap that would have produced a wrong answer confidently: geocoded permit points that never land inside a polygon, and a permitting jurisdiction whose records the county does not hold. Where a public registry is used to validate a public-data model, the registry’s own coverage has to be measured before its disagreement is read as model error.

Sites, not polygons.

At 21 cm the detector splits a roof into 1.80 polygons per site on average, against 1.30 for the earlier 0.6 m stack, so polygon counts are not comparable across resolutions and per-polygon downstream calculations overcount. The 3,362 detected polygons form 1,865 sites, and any downstream quantity that is naturally per roof should be computed per site.

5.1 Limitations

- **One county, one imagery vintage.** All detection results are on a 14.93 km² area of Santa Cruz County in 2025 imagery. The gate has not been re-run elsewhere, and outside the county’s imagery service the shipped model has no gated input at this resolution.
- **Resolution left on the table.** The county serves 6.3 cm imagery and we requested 21 cm, so these results are not a ceiling for the method.
- **The label-noise floor is unmeasured.** Without inter-annotator agreement we cannot say whether mask IoU of 0.844 is near what human labels allow.
- **Permit recall is era-restricted.** 73.6% holds for installs the imagery could contain; across all permit years it is 55.3%. Parcel matching can also credit a spurious detection that happens to fall on a permitted parcel; gate precision of 0.850 bounds that effect without removing it.
- **Tilt validation is distributional.** No filing can be matched to an array, so agreement is shown for the median only. The standoff experiment is per-array but rests on small groups ($n = 37$ and 54).

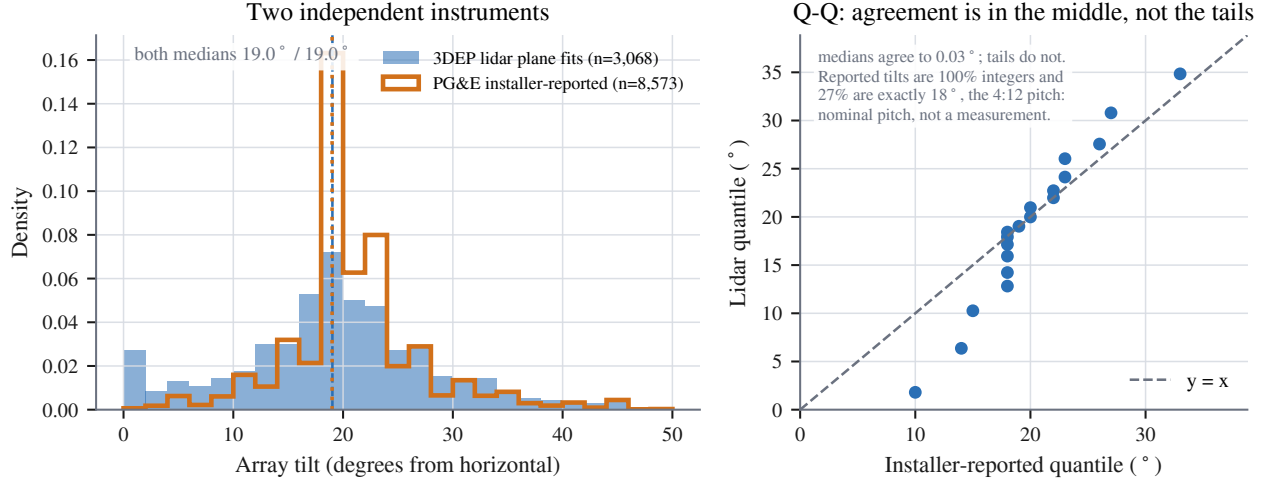


Figure 3: Per-array tilt from public lidar against installer-reported tilt from utility interconnection filings. The medians agree to 0.03°; the distributions do not, and the reason is reporting granularity rather than measurement error.

- **Two unexplained shifts.** The shipped model’s threshold-tuning penalty and chip-to-tile gap both grew relative to the earlier checkpoint. Neither affects the gate verdict, and neither is yet understood.

6 Conclusion

On public imagery, public lidar and permissively licensed models, rooftop arrays can be located, measured and oriented well enough to feed a per-roof analysis: tile-level F_1 0.826 under a gate that cannot be quietly relaxed, 73.6% recall against permits the model never saw, unbiased area once the mask prompt is sized correctly, and tilt whose central tendency matches installer records once a projection trap is removed. The detector is not the hard part of a public-data rooftop pipeline. The companion paper shows what is: public soiling labels are measured at stations, not roofs, and no quality of detection repairs that.

Code and data availability

Code, configuration, the gate script and the figure scripts are public in the repository named on page 1. The imagery tiles, labels and model weights are not redistributed there; the imagery and lidar are public services that the code fetches directly.

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